

=====

PROCESSING SECTOR: XLK

=====

--- Granger Causality: Macro -> XLK ---

[CPI\_Surprise]

Lag 1: p-value = 0.0207 \*\*

Lag 2: p-value = 0.1753

Lag 3: p-value = 0.4097

Lag 4: p-value = 0.7066

Lag 5: p-value = 0.8356

[PPI\_Surprise]

Lag 1: p-value = 0.0002 \*\*\*

Lag 2: p-value = 0.0125 \*\*

Lag 3: p-value = 0.0637 \*

Lag 4: p-value = 0.2301

Lag 5: p-value = 0.5242

[FedFunds\_Diff]

Lag 1: p-value = 0.0180 \*\*

Lag 2: p-value = 0.0000 \*\*\*

Lag 3: p-value = 0.0001 \*\*\*

Lag 4: p-value = 0.0002 \*\*\*

Lag 5: p-value = 0.0002 \*\*\*

Fitting VAR Model for XLK...

2026-06-21 22:57:35 - src.features - INFO - Engineering features and macro surprises for XLE...

2026-06-21 22:57:35 - src.features - INFO - Calculating Autoregressive Macro Surprises...

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#### LAG SELECTION CRITERIA

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VAR Order Selection (\* highlights the minimums)

=====

|  | AIC | BIC | FPE | HQIC |
|--|-----|-----|-----|------|
|--|-----|-----|-----|------|

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|   |         |         |            |         |
|---|---------|---------|------------|---------|
| 0 | -19.24  | -19.23  | 4.419e-09  | -19.24  |
| 1 | -24.42  | -24.39  | 2.492e-11  | -24.41  |
| 2 | -25.09  | -25.05* | 1.273e-11  | -25.07  |
| 3 | -25.09  | -25.04  | 1.267e-11  | -25.07  |
| 4 | -25.10  | -25.03  | 1.252e-11  | -25.08  |
| 5 | -25.12  | -25.03  | 1.236e-11  | -25.09  |
| 6 | -25.13  | -25.03  | 1.219e-11  | -25.09  |
| 7 | -25.14  | -25.02  | 1.210e-11  | -25.10  |
| 8 | -25.15  | -25.01  | 1.201e-11  | -25.10  |
| 9 | -25.17* | -25.01  | 1.175e-11* | -25.11* |

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## PROCESSING SECTOR: XLE

=====

--- Granger Causality: Macro -> XLE ---

[CPI\_Surprise]

Lag 1: p-value = 0.0119 \*\*

Lag 2: p-value = 0.0226 \*\*

Lag 3: p-value = 0.0871 \*

Lag 4: p-value = 0.3392

Lag 5: p-value = 0.5300

[PPI\_Surprise]

Lag 1: p-value = 0.0000 \*\*\*

Lag 2: p-value = 0.0001 \*\*\*

Lag 3: p-value = 0.0009 \*\*\*

Lag 4: p-value = 0.0229 \*\*

Lag 5: p-value = 0.1467

[FedFunds\_Diff]

Lag 1: p-value = 0.9799

Lag 2: p-value = 0.2404

Lag 3: p-value = 0.5514

Lag 4: p-value = 0.3750

Lag 5: p-value = 0.0051 \*\*\*

Fitting VAR Model for XLE...

2026-06-21 22:57:38 - src.features - INFO - Engineering features and macro surprises for XLF...

2026-06-21 22:57:38 - src.features - INFO - Calculating Autoregressive Macro Surprises...

=====

PROCESSING SECTOR: XLF

=====

--- Granger Causality: Macro -> XLF ---

[CPI\_Surprise]

Lag 1: p-value = 0.0000 \*\*\*

Lag 2: p-value = 0.0000 \*\*\*

Lag 3: p-value = 0.0000 \*\*\*

Lag 4: p-value = 0.0000 \*\*\*

Lag 5: p-value = 0.0000 \*\*\*

[PPI\_Surprise]

Lag 1: p-value = 0.0000 \*\*\*

Lag 2: p-value = 0.0000 \*\*\*

Lag 3: p-value = 0.0000 \*\*\*

Lag 4: p-value = 0.0000 \*\*\*

Lag 5: p-value = 0.0000 \*\*\*

[FedFunds\_Diff]

Lag 1: p-value = 0.3611

Lag 2: p-value = 0.1928

Lag 3: p-value = 0.0507 \*

Lag 4: p-value = 0.4022

Lag 5: p-value = 0.0911 \*

Fitting VAR Model for XLF...

2026-06-21 22:57:42 - src.features - INFO - Engineering features and macro surprises for XLV...

2026-06-21 22:57:42 - src.features - INFO - Calculating Autoregressive Macro Surprises...

=====

PROCESSING SECTOR: XLV

=====

--- Granger Causality: Macro -> XLV ---

[CPI\_Surprise]

Lag 1: p-value = 0.0715 \*

Lag 2: p-value = 0.4637

Lag 3: p-value = 0.4946

Lag 4: p-value = 0.6488

Lag 5: p-value = 0.8507

[PPI\_Surprise]

Lag 1: p-value = 0.0001 \*\*\*

Lag 2: p-value = 0.0432 \*\*

Lag 3: p-value = 0.0688 \*

Lag 4: p-value = 0.1218

Lag 5: p-value = 0.3092

[FedFunds\_Diff]

Lag 1: p-value = 0.1174

Lag 2: p-value = 0.0009 \*\*\*

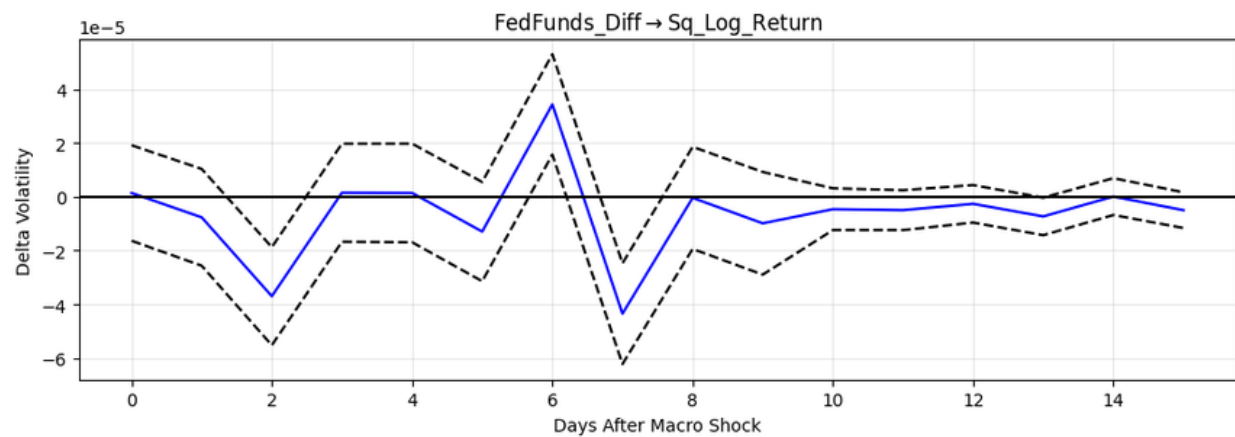
Lag 3: p-value = 0.0055 \*\*\*

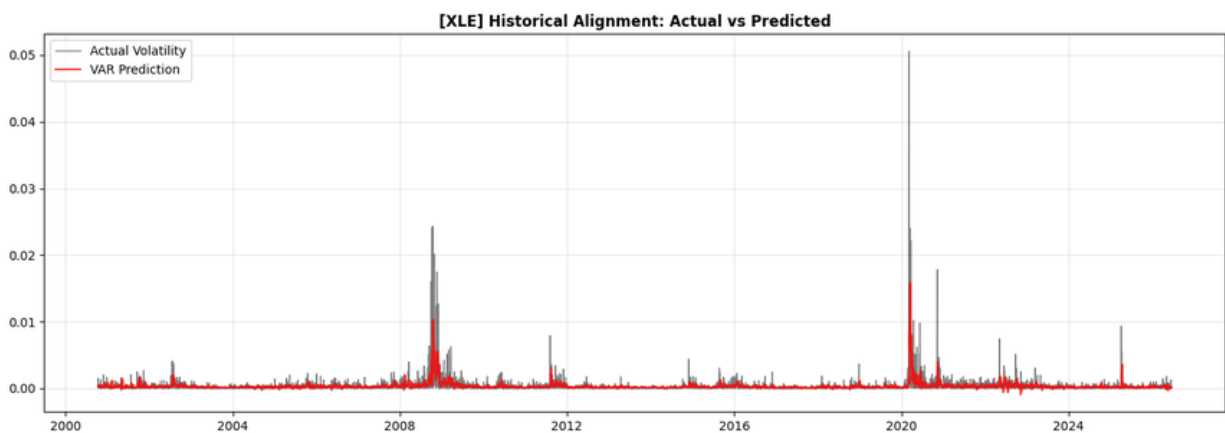
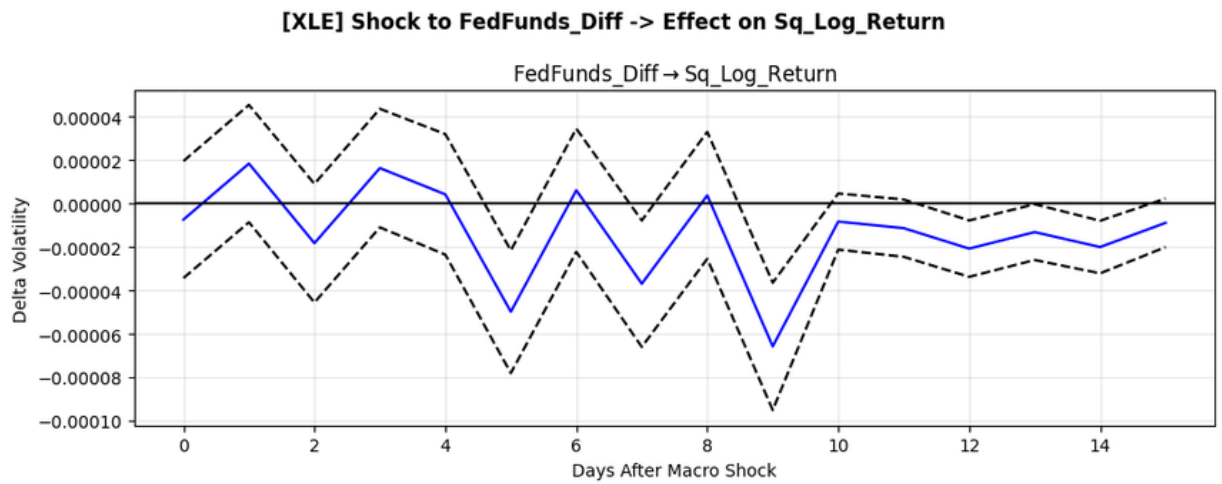
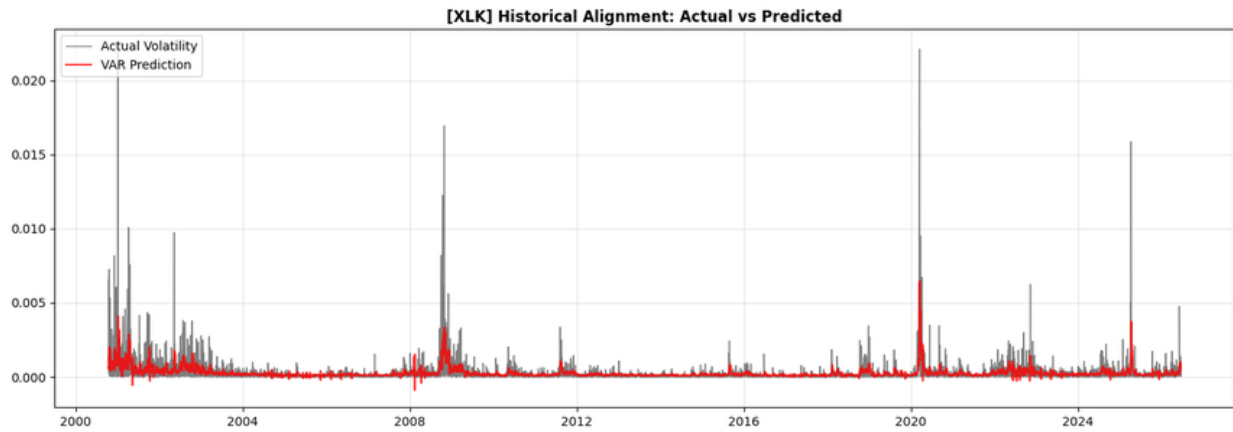
Lag 4: p-value = 0.0135 \*\*

Lag 5: p-value = 0.0001 \*\*\*

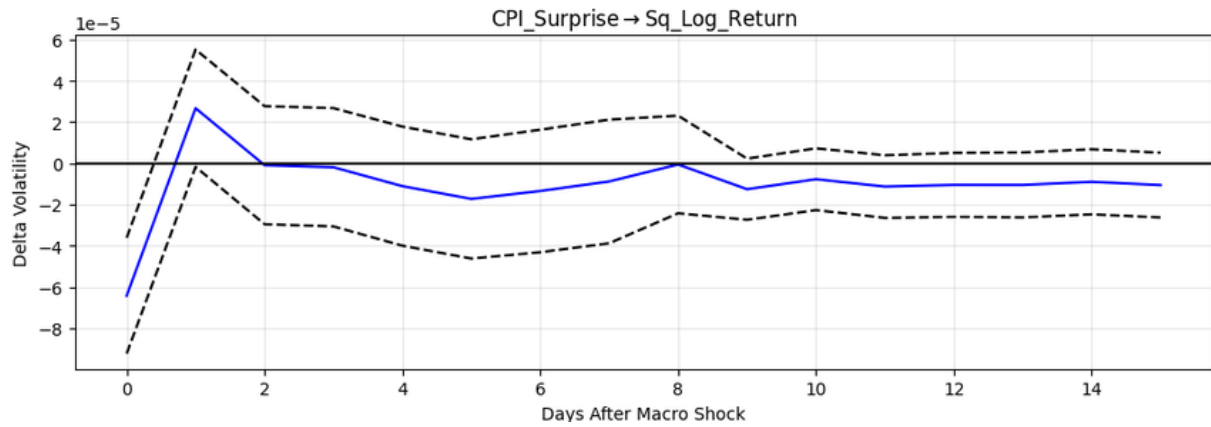
|                         | XLK     |        |     | XLE     |        |     | XLF     |        |     | XLV     |        |     |
|-------------------------|---------|--------|-----|---------|--------|-----|---------|--------|-----|---------|--------|-----|
|                         | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig |
| <b>const</b>            | 0.0001  | 0.0000 | *** | 0.0001  | 0.0000 | *** | 0.0001  | 0.0000 | *** | 0.0000  | 0.0000 | *** |
| <b>L1.CPI_Surprise</b>  | -0.0000 | 0.5638 |     | 0.0001  | 0.3700 |     | 0.0002  | 0.0138 | **  | -0.0000 | 0.7941 |     |
| <b>L2.CPI_Surprise</b>  | 0.0000  | 0.6436 |     | -0.0002 | 0.1204 |     | -0.0002 | 0.0985 | *   | 0.0000  | 0.3245 |     |
| <b>L2.FedFunds_Diff</b> | -0.0007 | 0.0001 | *** | -0.0003 | 0.2183 |     | -0.0001 | 0.6065 |     | -0.0003 | 0.0009 | *** |
| <b>L5.FedFunds_Diff</b> | -0.0002 | 0.1789 |     | -0.0010 | 0.0001 | *** | -0.0004 | 0.2086 |     | -0.0003 | 0.0002 | *** |
| <b>L6.FedFunds_Diff</b> | 0.0008  | 0.0000 | *** | 0.0001  | 0.7784 |     | 0.0002  | 0.5663 |     | 0.0000  | 0.8988 |     |
| <b>L7.FedFunds_Diff</b> | -0.0008 | 0.0000 | *** | -0.0006 | 0.0338 | **  | -0.0001 | 0.6908 |     | -0.0006 | 0.0000 | *** |
| <b>L8.FedFunds_Diff</b> | 0.0001  | 0.6171 |     | 0.0000  | 0.9107 |     | 0.0001  | 0.6408 |     | 0.0002  | 0.0348 | **  |
| <b>L9.FedFunds_Diff</b> | -0.0001 | 0.7399 |     | -0.0010 | 0.0002 | *** | -0.0003 | 0.3109 |     | -0.0003 | 0.0001 | *** |

[XLK] Shock to FedFunds\_Diff -> Effect on Sq\_Log\_Return

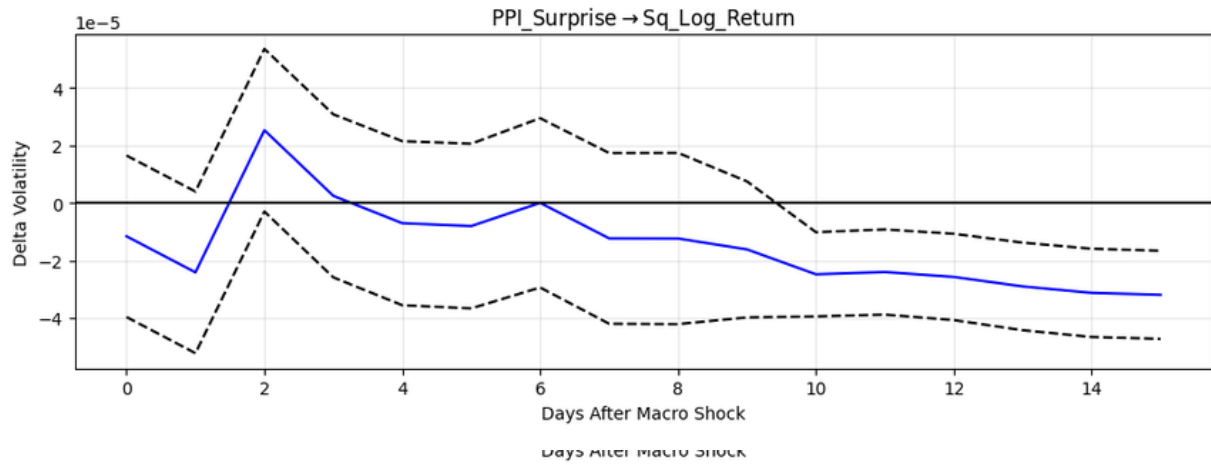




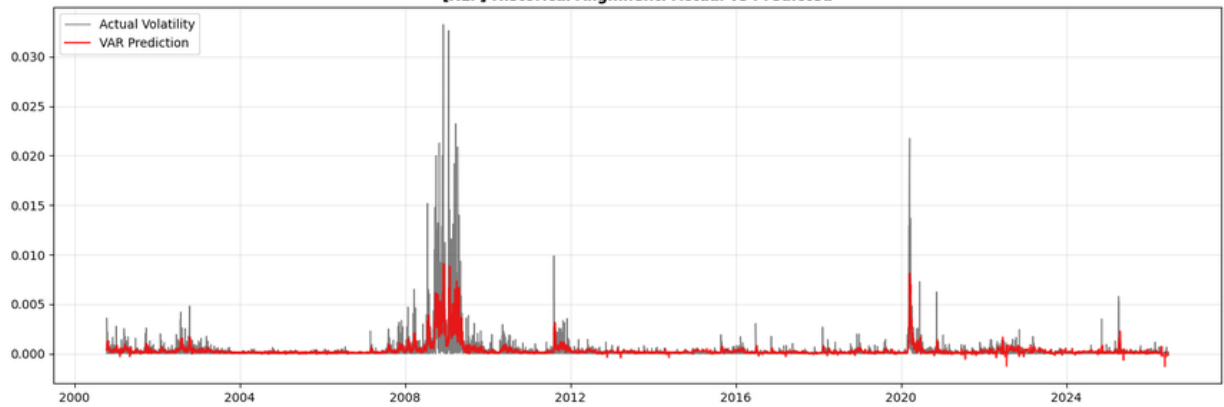
### [XLF] Shock to CPI\_Surprise -> Effect on Sq\_Log\_Return

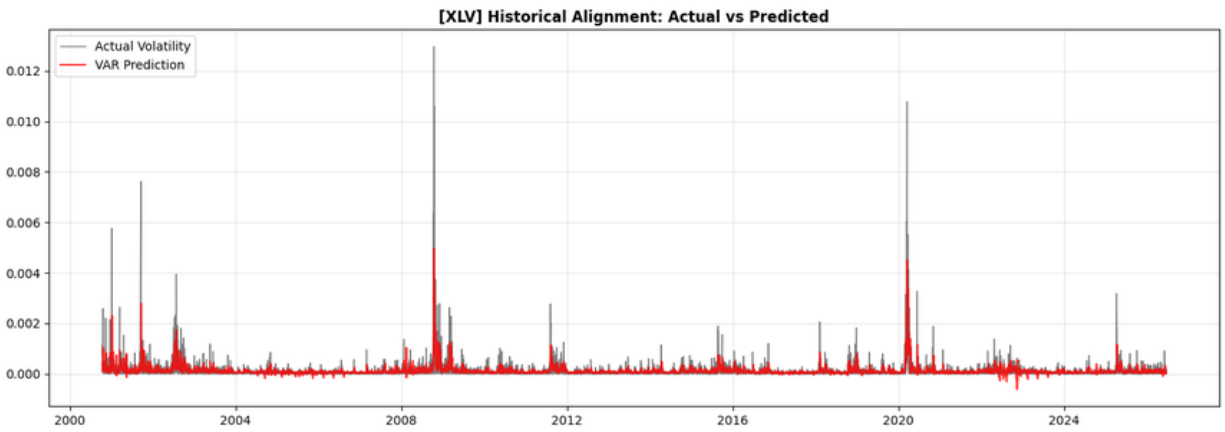
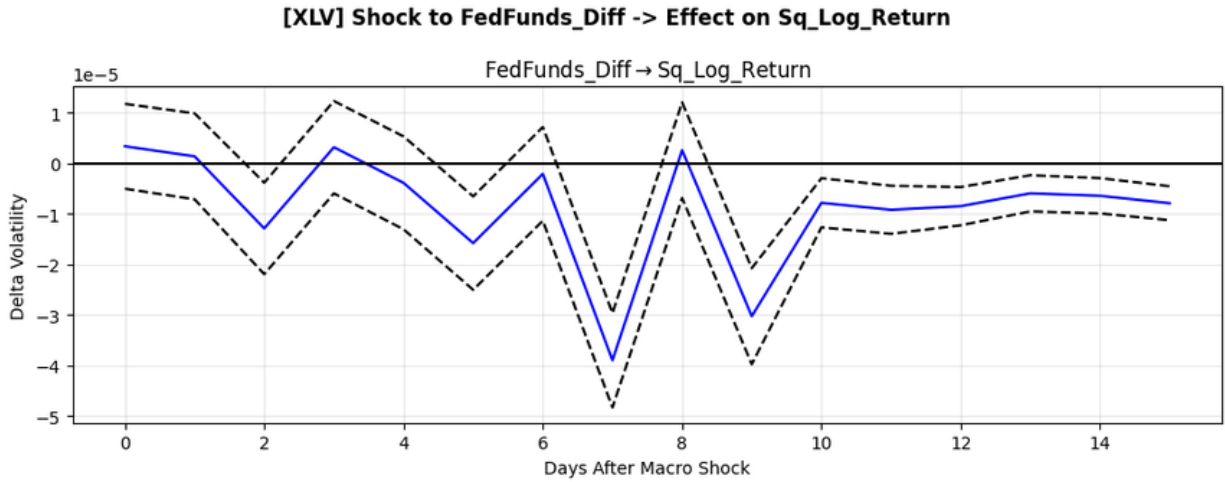


### [XLF] Shock to PPI\_Surprise -> Effect on Sq\_Log\_Return



### [XLF] Historical Alignment: Actual vs Predicted





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PROCESSING SECTOR: XLK

=====

--- Granger Causality: Macro -> XLK ---

[Unemployment\_Surprise]

Lag 1: p-value = 0.1338

Lag 2: p-value = 0.4742

Lag 3: p-value = 0.6605

Lag 4: p-value = 0.5588

Lag 5: p-value = 0.9252

Lag 6: p-value = 0.9444

Lag 7: p-value = 0.9560

Lag 8: p-value = 0.9450

Lag 9: p-value = 0.8940

[Housing\_Starts\_Surprise]

Lag 1: p-value = 0.0084 \*\*\*

Lag 2: p-value = 0.1010

Lag 3: p-value = 0.2792

Lag 4: p-value = 0.5451

Lag 5: p-value = 0.7538

Lag 6: p-value = 0.8051

Lag 7: p-value = 0.8902

Lag 8: p-value = 0.9517

Lag 9: p-value = 0.9785

Fitting VAR Model for XLK...

2026-06-22 15:57:02 - src.features - INFO - Engineering features and macro surprises for XLE...

2026-06-22 15:57:02 - src.features - INFO - Calculating Autoregressive Macro Surprises...

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LAG SELECTION CRITERIA

=====

VAR Order Selection (\* highlights the minimums)

=====

|  | AIC | BIC | FPE | HQIC |
|--|-----|-----|-----|------|
|--|-----|-----|-----|------|

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|   |         |         |            |         |
|---|---------|---------|------------|---------|
| 0 | -6.105  | -6.101  | 0.002233   | -6.104  |
| 1 | -11.18  | -11.17  | 1.397e-05  | -11.17  |
| 2 | -11.23  | -11.21  | 1.323e-05  | -11.23  |
| 3 | -11.24  | -11.21  | 1.314e-05  | -11.23  |
| 4 | -11.25  | -11.21  | 1.294e-05  | -11.24  |
| 5 | -11.27  | -11.22  | 1.280e-05  | -11.25  |
| 6 | -11.28  | -11.22* | 1.262e-05  | -11.26  |
| 7 | -11.28  | -11.21  | 1.262e-05  | -11.26  |
| 8 | -11.29  | -11.21  | 1.247e-05  | -11.26* |
| 9 | -11.30* | -11.21  | 1.243e-05* | -11.26  |

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PROCESSING SECTOR: XLE

=====

--- Granger Causality: Macro -> XLE ---

[Unemployment\_Surprise]

Lag 1: p-value = 0.0033 \*\*\*

Lag 2: p-value = 0.0907 \*

Lag 3: p-value = 0.2890

Lag 4: p-value = 0.1681  
Lag 5: p-value = 0.1502  
Lag 6: p-value = 0.1859  
Lag 7: p-value = 0.0281 \*\*  
Lag 8: p-value = 0.0520 \*  
Lag 9: p-value = 0.0498 \*\*

[Housing\_Starts\_Surprise]

Lag 1: p-value = 0.0096 \*\*\*  
Lag 2: p-value = 0.1130  
Lag 3: p-value = 0.3197  
Lag 4: p-value = 0.6612  
Lag 5: p-value = 0.8758  
Lag 6: p-value = 0.9319  
Lag 7: p-value = 0.9589  
Lag 8: p-value = 0.9674  
Lag 9: p-value = 0.9808

Fitting VAR Model for XLE...

2026-06-22 15:57:04 - src.features - INFO - Engineering features and macro surprises for XLF...

2026-06-22 15:57:04 - src.features - INFO - Calculating Autoregressive Macro Surprises...

=====

PROCESSING SECTOR: XLF

=====

--- Granger Causality: Macro -> XLF ---

[Unemployment\_Surprise]

Lag 1: p-value = 0.0687 \*

Lag 2: p-value = 0.2403

Lag 3: p-value = 0.6530

Lag 4: p-value = 0.5815

Lag 5: p-value = 0.6280

Lag 6: p-value = 0.9091

Lag 7: p-value = 0.9395

Lag 8: p-value = 0.9522

Lag 9: p-value = 0.9741

[Housing\_Starts\_Surprise]

Lag 1: p-value = 0.0000 \*\*\*

Lag 2: p-value = 0.0000 \*\*\*

Lag 3: p-value = 0.0000 \*\*\*

Lag 4: p-value = 0.0003 \*\*\*

Lag 5: p-value = 0.0034 \*\*\*

Lag 6: p-value = 0.0615 \*

Lag 7: p-value = 0.1561

Lag 8: p-value = 0.2314

Lag 9: p-value = 0.2302

Fitting VAR Model for XLF...

2026-06-22 15:57:06 - src.features - INFO - Engineering features and macro surprises for XLV...

2026-06-22 15:57:06 - src.features - INFO - Calculating Autoregressive Macro Surprises...

=====

PROCESSING SECTOR: XLV

=====

--- Granger Causality: Macro -> XLV ---

[Unemployment\_Surprise]

Lag 1: p-value = 0.2564

Lag 2: p-value = 0.3490

Lag 3: p-value = 0.1599

Lag 4: p-value = 0.1289

Lag 5: p-value = 0.3265

Lag 6: p-value = 0.3644

Lag 7: p-value = 0.4593

Lag 8: p-value = 0.4893

Lag 9: p-value = 0.4407

[Housing\_Starts\_Surprise]

Lag 1: p-value = 0.0493 \*\*

Lag 2: p-value = 0.4397

Lag 3: p-value = 0.7300

Lag 4: p-value = 0.8572

Lag 5: p-value = 0.9475

Lag 6: p-value = 0.8157

Lag 7: p-value = 0.8969

Lag 8: p-value = 0.9497

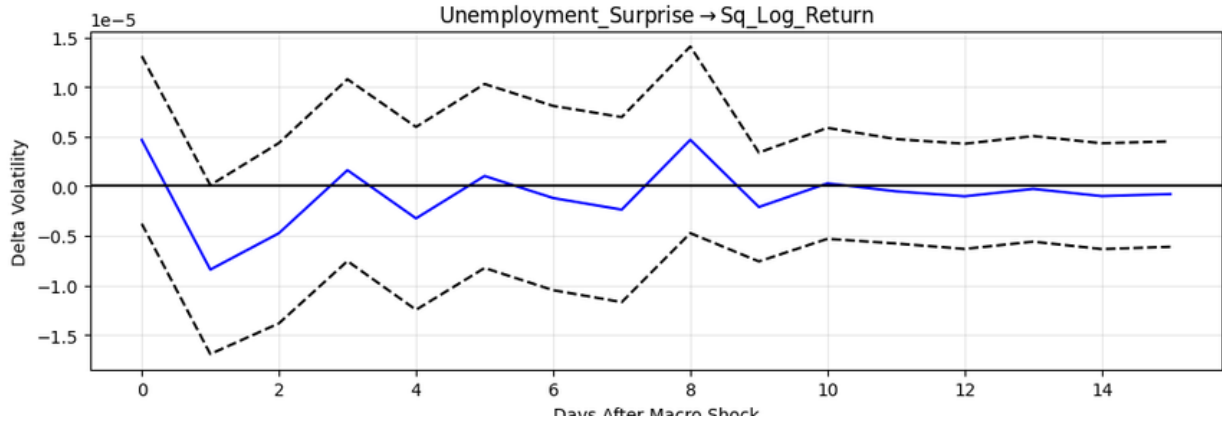
Lag 9: p-value = 0.9731

|                                 | XLK     |        |     | XLE     |        |     | XLF     |        |     | XLV  |        |     |
|---------------------------------|---------|--------|-----|---------|--------|-----|---------|--------|-----|------|--------|-----|
|                                 | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig | Coef | P-Val  | Sig |
| <b>const</b>                    | 0.0001  | 0.0000 | *** | 0.0001  | 0.0000 | *** | 0.0001  | 0.0000 | *** | 0.0  | 0.0000 | *** |
| <b>L1.Unemployment_Surprise</b> | -0.0000 | 0.6600 |     | -0.0001 | 0.4536 |     | 0.0001  | 0.4584 |     | -0.0 | 0.0359 | **  |
| <b>L3.Unemployment_Surprise</b> | 0.0001  | 0.3600 |     | -0.0001 | 0.2230 |     | -0.0001 | 0.5440 |     | 0.0  | 0.0947 | *   |
| <b>L4.Unemployment_Surprise</b> | -0.0000 | 0.4636 |     | 0.0002  | 0.0095 | *** | 0.0001  | 0.3916 |     | -0.0 | 0.3273 |     |
| <b>L7.Unemployment_Surprise</b> | 0.0001  | 0.3993 |     | -0.0002 | 0.0964 | *   | -0.0000 | 0.8051 |     | -0.0 | 0.7606 |     |

**[XLE] Shock to Unemployment\_Surprise -> Effect on Sq\_Log\_Return**



**[XLV] Shock to Unemployment\_Surprise -> Effect on Sq\_Log\_Return**



=====

PROCESSING SECTOR: XLK

=====

--- Granger Causality: Macro -> XLK ---

[Trade\_Balance\_Surprise]

Lag 1: p-value = 0.0426 \*\*

Lag 2: p-value = 0.2843

Lag 3: p-value = 0.5691

Lag 4: p-value = 0.6897

Lag 5: p-value = 0.8466

Lag 6: p-value = 0.9107

Lag 7: p-value = 0.9500

Lag 8: p-value = 0.9770

Lag 9: p-value = 0.9891

[M1\_Money\_Surprise]

Lag 1: p-value = 0.5391

Lag 2: p-value = 0.8782

Lag 3: p-value = 0.9509

Lag 4: p-value = 0.9862

Lag 5: p-value = 0.9960

Lag 6: p-value = 0.9993

Lag 7: p-value = 0.9998

Lag 8: p-value = 1.0000

Lag 9: p-value = 0.9333

Fitting VAR Model for XLK...

2026-06-22 15:56:28 - src.features - INFO - Engineering features and macro surprises for XLE...

2026-06-22 15:56:28 - src.features - INFO - Calculating Autoregressive Macro Surprises...

=====

#### LAG SELECTION CRITERIA

=====

VAR Order Selection (\* highlights the minimums)

=====

|       | AIC    | BIC    | FPE        | HQIC   |
|-------|--------|--------|------------|--------|
| ----- |        |        |            |        |
| 0     | 17.26  | 17.26  | 3.129e+07  | 17.26  |
| 1     | 12.23  | 12.25  | 2.055e+05  | 12.24  |
| 2     | 12.18  | 12.21  | 1.957e+05  | 12.19  |
| 3     | 12.18  | 12.21  | 1.945e+05  | 12.19  |
| 4     | 12.16  | 12.20  | 1.915e+05  | 12.18  |
| 5     | 12.15  | 12.20  | 1.893e+05  | 12.17  |
| 6     | 12.13  | 12.19* | 1.857e+05  | 12.15  |
| 7     | 12.13  | 12.20  | 1.857e+05  | 12.16  |
| 8     | 12.12  | 12.20  | 1.837e+05  | 12.15* |
| 9     | 12.12* | 12.21  | 1.833e+05* | 12.15  |

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PROCESSING SECTOR: XLE

=====

--- Granger Causality: Macro -> XLE ---

[Trade\_Balance\_Surprise]

Lag 1: p-value = 0.0097 \*\*\*

Lag 2: p-value = 0.1338

Lag 3: p-value = 0.3819

Lag 4: p-value = 0.7187

Lag 5: p-value = 0.8828

Lag 6: p-value = 0.9405

Lag 7: p-value = 0.9841

Lag 8: p-value = 0.8935

Lag 9: p-value = 0.9231

[M1\_Money\_Surprise]

Lag 1: p-value = 0.2015

Lag 2: p-value = 0.6341

Lag 3: p-value = 0.8627

Lag 4: p-value = 0.9271

Lag 5: p-value = 0.6712

Lag 6: p-value = 0.7706

Lag 7: p-value = 0.6916

Lag 8: p-value = 0.6629

Lag 9: p-value = 0.4240

Fitting VAR Model for XLE...

2026-06-22 15:56:31 - src.features - INFO - Engineering features and macro surprises for XLF...

2026-06-22 15:56:31 - src.features - INFO - Calculating Autoregressive Macro Surprises...

=====

PROCESSING SECTOR: XLF

=====

--- Granger Causality: Macro -> XLF ---

[Trade\_Balance\_Surprise]

Lag 1: p-value = 0.0000 \*\*\*

Lag 2: p-value = 0.0000 \*\*\*

Lag 3: p-value = 0.0000 \*\*\*

Lag 4: p-value = 0.0009 \*\*\*

Lag 5: p-value = 0.0061 \*\*\*

Lag 6: p-value = 0.0996 \*

Lag 7: p-value = 0.2317

Lag 8: p-value = 0.3200

Lag 9: p-value = 0.0357 \*\*

[M1\_Money\_Surprise]

Lag 1: p-value = 0.0946 \*

Lag 2: p-value = 0.3662

Lag 3: p-value = 0.6340

Lag 4: p-value = 0.8294

Lag 5: p-value = 0.9325

Lag 6: p-value = 0.9852

Lag 7: p-value = 0.9958

Lag 8: p-value = 0.9987

Lag 9: p-value = 0.9846

Fitting VAR Model for XLF...

2026-06-22 15:56:33 - src.features - INFO - Engineering features and macro surprises for XLV...

2026-06-22 15:56:33 - src.features - INFO - Calculating Autoregressive Macro Surprises...

=====

PROCESSING SECTOR: XLV

=====

--- Granger Causality: Macro -> XLV ---

[Trade\_Balance\_Surprise]

Lag 1: p-value = 0.0041 \*\*\*

Lag 2: p-value = 0.1750

Lag 3: p-value = 0.3755

Lag 4: p-value = 0.2129

Lag 5: p-value = 0.3689

Lag 6: p-value = 0.3921

Lag 7: p-value = 0.4733

Lag 8: p-value = 0.6292

Lag 9: p-value = 0.7196

[M1\_Money\_Surprise]

Lag 1: p-value = 0.1930

Lag 2: p-value = 0.5330

Lag 3: p-value = 0.7735

Lag 4: p-value = 0.8877

Lag 5: p-value = 0.9512

Lag 6: p-value = 0.9756

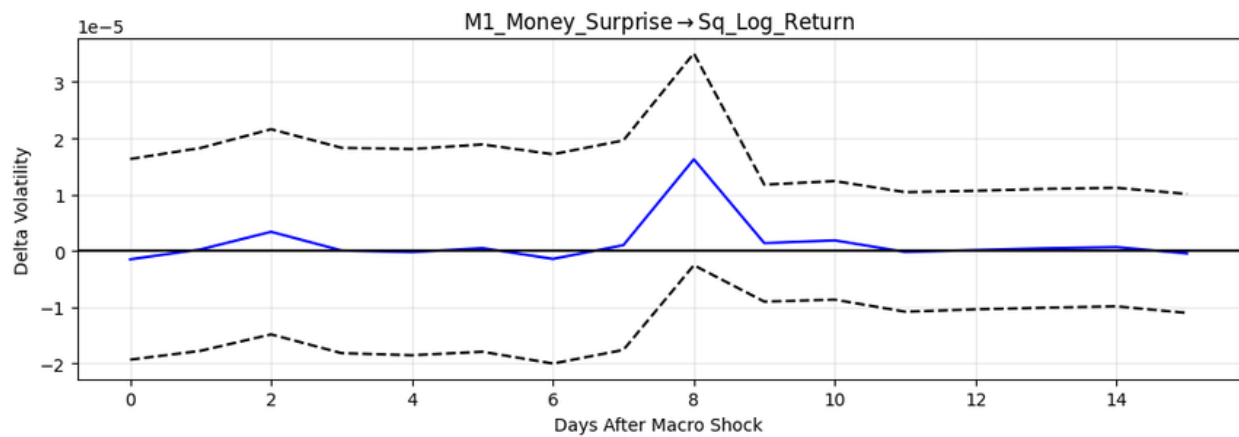
Lag 7: p-value = 0.9913

Lag 8: p-value = 0.9975

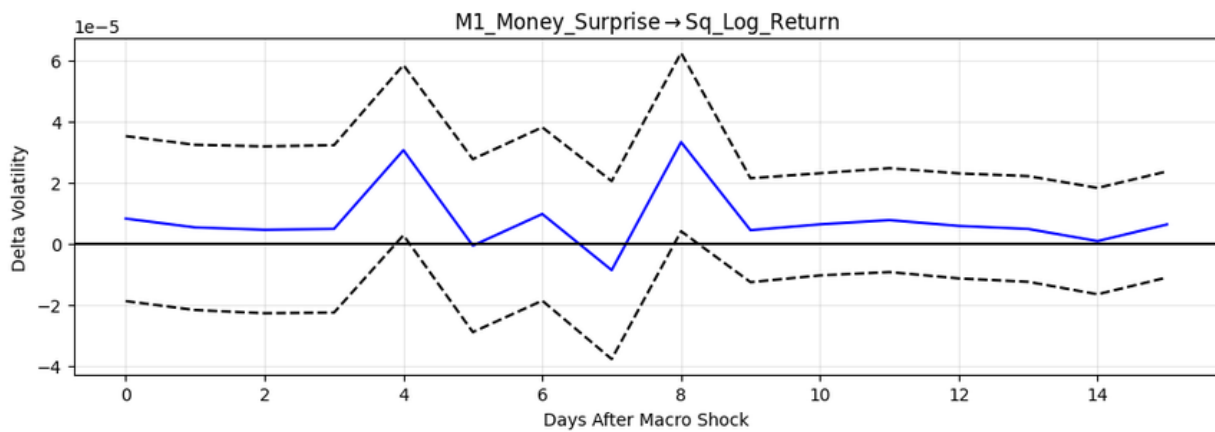
Lag 9: p-value = 0.2653

|                           | XLK     |        |     | XLE     |        |     | XLF     |        |     | XLV  |        |     |
|---------------------------|---------|--------|-----|---------|--------|-----|---------|--------|-----|------|--------|-----|
|                           | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig | Coef | P-Val  | Sig |
| const                     | 0.0001  | 0.0000 | *** | 0.0001  | 0.0000 | *** | 0.0001  | 0.0000 | *** | 0.0  | 0.0000 | *** |
| L3.Trade_Balance_Surprise | 0.0000  | 0.5575 |     | 0.0000  | 0.3742 |     | 0.0000  | 0.4826 |     | 0.0  | 0.0867 | *   |
| L8.Trade_Balance_Surprise | -0.0000 | 0.9624 |     | 0.0000  | 0.1965 |     | 0.0000  | 0.0141 | **  | 0.0  | 0.9320 |     |
| L8.M1_Money_Surprise      | 0.0000  | 0.2095 |     | 0.0000  | 0.0373 | **  | 0.0000  | 0.3524 |     | 0.0  | 0.0129 | **  |
| L9.Trade_Balance_Surprise | 0.0000  | 0.9342 |     | -0.0000 | 0.7385 |     | -0.0000 | 0.0018 | *** | -0.0 | 0.8296 |     |
| L9.M1_Money_Surprise      | -0.0000 | 0.0559 | *   | -0.0000 | 0.0643 | *   | -0.0000 | 0.1914 |     | -0.0 | 0.0009 | *** |

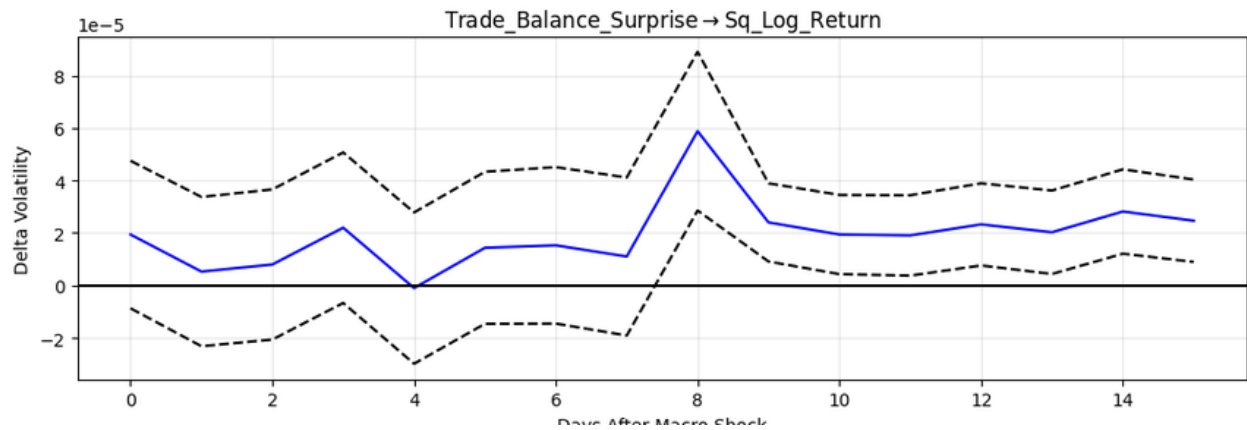
[XLK] Shock to M1\_Money\_Surprise -> Effect on Sq\_Log\_Return



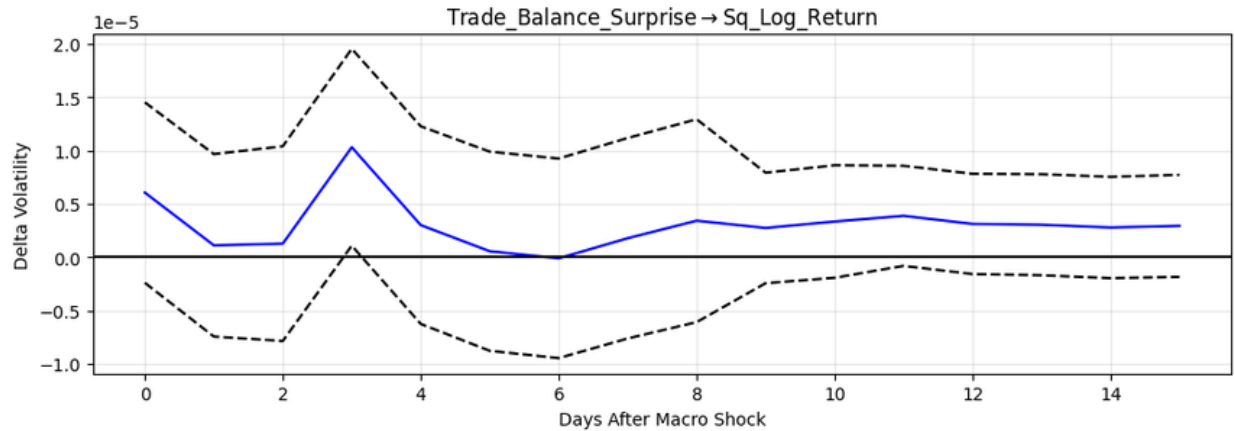
[XLE] Shock to M1\_Money\_Surprise -> Effect on Sq\_Log\_Return



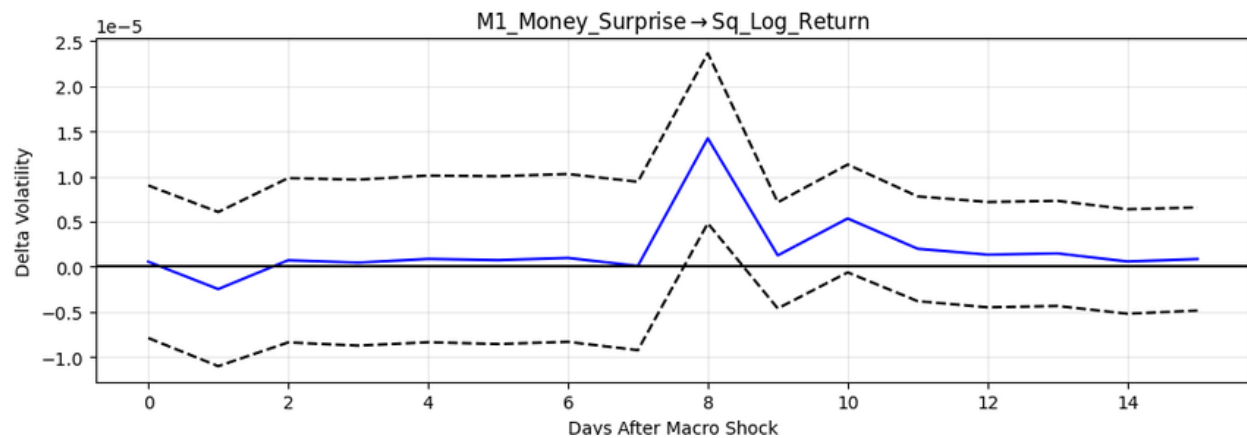
**[XLF] Shock to Trade\_Balance\_Surprise -> Effect on Sq\_Log\_Return**



**[XLV] Shock to Trade\_Balance\_Surprise -> Effect on Sq\_Log\_Return**



**[XLV] Shock to M1\_Money\_Surprise -> Effect on Sq\_Log\_Return**



=====

PROCESSING SECTOR: XLK

=====

--- Granger Causality: Macro -> XLK ---

[NFP\_Surprise]

Lag 1: p-value = 0.0301 \*\*

Lag 2: p-value = 0.2076

Lag 3: p-value = 0.3365

Lag 4: p-value = 0.3065

Lag 5: p-value = 0.7409

Lag 6: p-value = 0.8400

Lag 7: p-value = 0.8715

Lag 8: p-value = 0.8728

Lag 9: p-value = 0.8407

[Retail\_Sales\_Surprise]

Lag 1: p-value = 0.0075 \*\*\*

Lag 2: p-value = 0.1033

Lag 3: p-value = 0.2805

Lag 4: p-value = 0.5879

Lag 5: p-value = 0.8269

Lag 6: p-value = 0.9634

Lag 7: p-value = 0.9837

Lag 8: p-value = 0.9912

Lag 9: p-value = 0.9973

Fitting VAR Model for XLK...

2026-06-22 15:56:14 - src.features - INFO - Engineering features and macro surprises for XLE...

2026-06-22 15:56:14 - src.features - INFO - Calculating Autoregressive Macro Surprises...

=====

#### LAG SELECTION CRITERIA

=====

VAR Order Selection (\* highlights the minimums)

=====

|       | AIC    | BIC    | FPE        | HQIC   |
|-------|--------|--------|------------|--------|
| ----- |        |        |            |        |
| 0     | 18.33  | 18.33  | 9.128e+07  | 18.33  |
| 1     | 13.68  | 13.69  | 8.694e+05  | 13.68  |
| 2     | 13.62  | 13.64  | 8.183e+05  | 13.62  |
| 3     | 13.60  | 13.63  | 8.080e+05  | 13.61  |
| 4     | 13.58  | 13.62  | 7.895e+05  | 13.59  |
| 5     | 13.56  | 13.61  | 7.733e+05  | 13.58  |
| 6     | 12.31  | 12.37* | 2.230e+05  | 12.34  |
| 7     | 12.31  | 12.38  | 2.230e+05  | 12.34  |
| 8     | 12.30  | 12.38  | 2.205e+05  | 12.33* |
| 9     | 12.30* | 12.39  | 2.198e+05* | 12.33  |

-----

=====

PROCESSING SECTOR: XLE

=====

--- Granger Causality: Macro -> XLE ---

[NFP\_Surprise]

Lag 1: p-value = 0.0000 \*\*\*

Lag 2: p-value = 0.0011 \*\*\*

Lag 3: p-value = 0.0197 \*\*

Lag 4: p-value = 0.0287 \*\*

Lag 5: p-value = 0.0754 \*

Lag 6: p-value = 0.0763 \*

Lag 7: p-value = 0.0276 \*\*

Lag 8: p-value = 0.0495 \*\*

Lag 9: p-value = 0.0424 \*\*

[Retail\_Sales\_Surprise]

Lag 1: p-value = 0.0003 \*\*\*

Lag 2: p-value = 0.0410 \*\*

Lag 3: p-value = 0.1897

Lag 4: p-value = 0.4597

Lag 5: p-value = 0.8063

Lag 6: p-value = 0.8832

Lag 7: p-value = 0.9169

Lag 8: p-value = 0.9554

Lag 9: p-value = 0.9683

Fitting VAR Model for XLE...

2026-06-22 15:56:17 - src.features - INFO - Engineering features and macro surprises for XLF...

2026-06-22 15:56:17 - src.features - INFO - Calculating Autoregressive Macro Surprises...

=====

PROCESSING SECTOR: XLF

=====

--- Granger Causality: Macro -> XLF ---

[NFP\_Surprise]

Lag 1: p-value = 0.0000 \*\*\*

Lag 2: p-value = 0.0000 \*\*\*

Lag 3: p-value = 0.0001 \*\*\*

Lag 4: p-value = 0.0013 \*\*\*

Lag 5: p-value = 0.0086 \*\*\*

Lag 6: p-value = 0.1819

Lag 7: p-value = 0.2885

Lag 8: p-value = 0.3697

Lag 9: p-value = 0.5744

[Retail\_Sales\_Surprise]

Lag 1: p-value = 0.0000 \*\*\*

Lag 2: p-value = 0.0000 \*\*\*

Lag 3: p-value = 0.0000 \*\*\*

Lag 4: p-value = 0.0006 \*\*\*

Lag 5: p-value = 0.0072 \*\*\*

Lag 6: p-value = 0.1031

Lag 7: p-value = 0.0513 \*

Lag 8: p-value = 0.0771 \*

Lag 9: p-value = 0.1593

Fitting VAR Model for XLF...

2026-06-22 15:56:20 - src.features - INFO - Engineering features and macro surprises for XLV...

2026-06-22 15:56:20 - src.features - INFO - Calculating Autoregressive Macro Surprises...

=====

PROCESSING SECTOR: XLV

=====

--- Granger Causality: Macro -> XLV ---

[NFP\_Surprise]

Lag 1: p-value = 0.0053 \*\*\*

Lag 2: p-value = 0.1570

Lag 3: p-value = 0.0962 \*

Lag 4: p-value = 0.0609 \*

Lag 5: p-value = 0.2143

Lag 6: p-value = 0.2695

Lag 7: p-value = 0.3870

Lag 8: p-value = 0.4489

Lag 9: p-value = 0.4400

[Retail\_Sales\_Surprise]

Lag 1: p-value = 0.0209 \*\*

Lag 2: p-value = 0.4157

Lag 3: p-value = 0.4144

Lag 4: p-value = 0.5008

Lag 5: p-value = 0.7795

Lag 6: p-value = 0.8777

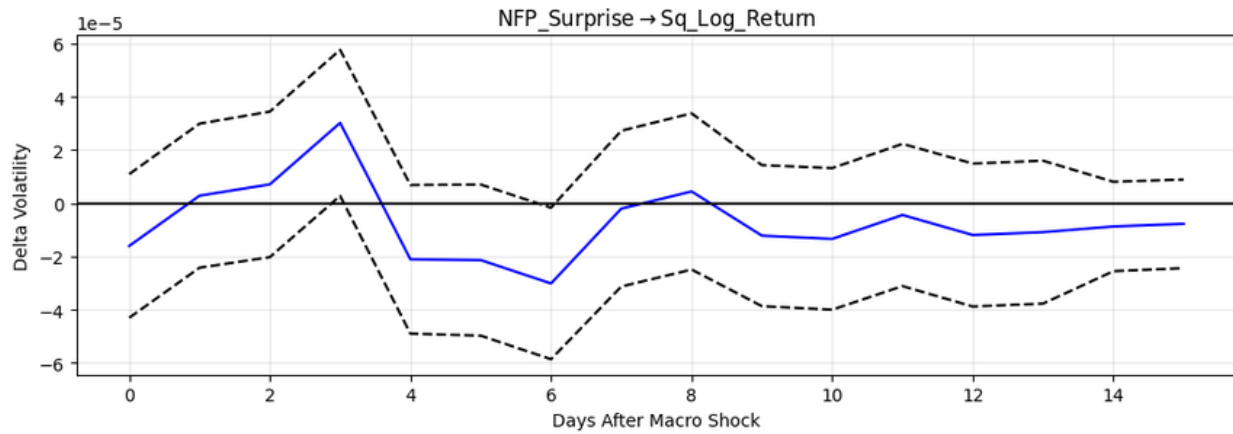
Lag 7: p-value = 0.9005

Lag 8: p-value = 0.9653

Lag 9: p-value = 0.9827

|                          | XLK     |        |     | XLE     |        |     | XLF     |        |     | XLV  |        |     |
|--------------------------|---------|--------|-----|---------|--------|-----|---------|--------|-----|------|--------|-----|
|                          | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig | Coef    | P-Val  | Sig | Coef | P-Val  | Sig |
| const                    | 0.0001  | 0.0000 | *** | 0.0001  | 0.0000 | *** | 0.0001  | 0.0000 | *** | 0.0  | 0.0000 | *** |
| L1.NFP_Surprise          | 0.0000  | 0.5550 |     | 0.0000  | 0.8012 |     | -0.0000 | 0.0640 | *   | 0.0  | 0.0995 | *   |
| L1.Retail_Sales_Surprise | -0.0000 | 0.3543 |     | 0.0000  | 0.0001 | *** | 0.0000  | 0.7810 |     | -0.0 | 0.7810 |     |
| L2.Retail_Sales_Surprise | 0.0000  | 0.2718 |     | -0.0000 | 0.0817 | *   | 0.0000  | 0.2987 |     | 0.0  | 0.3512 |     |
| L3.NFP_Surprise          | -0.0000 | 0.2495 |     | 0.0000  | 0.2457 |     | 0.0000  | 0.5073 |     | -0.0 | 0.0506 | *   |
| L4.NFP_Surprise          | 0.0000  | 0.4465 |     | -0.0000 | 0.0091 | *** | -0.0000 | 0.3614 |     | 0.0  | 0.2127 |     |
| L6.NFP_Surprise          | 0.0000  | 0.1865 |     | -0.0000 | 0.0012 | *** | 0.0000  | 0.8945 |     | 0.0  | 0.6485 |     |
| L7.NFP_Surprise          | -0.0000 | 0.1586 |     | 0.0000  | 0.0220 | **  | -0.0000 | 0.3260 |     | -0.0 | 0.5932 |     |
| L9.NFP_Surprise          | 0.0000  | 0.0793 | *   | -0.0000 | 0.9012 |     | 0.0000  | 0.5467 |     | 0.0  | 0.2422 |     |

**[XLE] Shock to NFP\_Surprise -> Effect on Sq\_Log\_Return**



### [XLE] Shock to Retail\_Sales\_Surprise -> Effect on Sq\_Log\_Return



### [XLF] Shock to Retail\_Sales\_Surprise -> Effect on Sq\_Log\_Return

